A-A-50798
April 30, 1982

COMMERCIAL ITEM DESCRIPTION

FREQUENCY STANDARD, OSCILLATOR (5MHz, 1MHz, and 100kHz)

The General Services Administration has authorized the use of this commercial item description.

This description covers a frequency standard hereinafter called equipment, with outputs of 5 MHz, 1 MHz and 100 kHz.

APPLICABLE DOCUMENTS

The following documents of the issue in effect on date of Invitations for Bids or Request for Proposal form a part of this Commercial Item Description to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-28777	Cable Assembly, Electronic Test Equipment (3 Wire, 3 Prong, 125 Volt, Molded) Grounding Plug Connectors
MIL-T-28800	Test Equipment for Use with Electrical and Electronic Equipment, General Specification For

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

SALIENT CHARACTERISTICS.

1 SOLID STATE CONSTRUCTION. The equipment shall be of solid state construction. Electron tubes shall not be used.

2. MAINFRAME PLUG-IN CONCEPT. Equipment utilizing externally accessible, externally removable subassemblies (mainframe plug-in concept) do not meet the requirements of this item description and therefore are unacceptable. Items using this concept may be offered, provided the plug-in(s) shall not be removable by access through the front panel or rear panel with the unaided hand. A system part number shall be assigned to any such mainframe plug-in combination required to meet the requirements of this specification. This system part number shall be marked on the mainframe identification plate and any plug-in(s).

3. ELECTRICAL POWER SOURCES AND CONNECTIONS.

a. POWER SOURCES.

(1) Alternating current. The equipment shall operate from the type III 50, 60, and 400 Hz single phase 115/230 volts power sources requirements of MIL-T-28800.

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(2) External dc. The equipment shall operate from an external 28 volts dc source.

(3) Batteries.

(a) Internal batteries. The equipment shall operate for at least 8 hours from internal batteries after warm-up from an external source.

(b) Battery charge. Equipment shall comply with the Type III battery charger requirements of MIL-T-28800.

b. MAXIMUM POWER.

The equipment shall consume not more than 70 watts in warm-up and 17 watts in operation.

c. POWER CONNECTIONS.

The equipment shall comply with the Type III power connnection requirements of MIL-T-28800 except as specified herein.

(1) Power cables.

(a) 115 Vac operation The power cables shall be in accordance with MIL-C-28777/3 except that the molded connector at the equipment end shall mate with Switchcraft part number EAC 302 connector or equal.

(b) 230 Vac operation The power cables shall be as specified in the contract or order

d POWER SELECTION SWITCH.

A power selection switch shall be provided as specified for Type III in MIL-T-28800 (115 Vac, 230 Vac).

4. DIMENSIONS AND WEIGHT.

a. DIMENSIONS. The dimensions shall not exceed 135 mm (5.3 inches) high by 447 mm (17 6 inches) wide by 362 mm (14 inches) deep.

b. WEIGHT. The maximum weight shall not exceed 14 kg (30 pounds).

5. PERFORMANCE REQUIREMENTS.

a. FREQUENCY OUTPUTS. Simultaneous 5 MHz, 1 MHz, and 100 kHz frequencies shall be available through separate connectors and shall be available on the front and rear panel. The frequency outputs shall be at least 1 volt RMS into 50 ohms.

b. CLOCK OUTPUT. A 1 MHz, 0.5 volt RMS into 1 kilohm, clock output shall be available on the rear panel.

c. FREQUENCY STABILITY.

(1) Aging rate. The aging rate shall be less than $\pm$ 5 x 10^{-10} per 24 hours.

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(2) Short term stability. For a 5 MHz output the short term stability shall be less than $\pm$ 1 x 10^{-11} for a 1 second averaging time with 60 separate samples being averaged.

(3) Temperature stability. The temperature stability shall be less than $\pm$ 2.5 x 10^{-9} for any temperature change within 0 degree C to 50 degrees C.

(4) Loading stability. The loading stability shall be less than $\pm$ 1 x 10^{-11} from open circuit to a 50 ohms load.

(5) Supply voltage stability. Supply voltage stability shall be less than $\pm$ 5 x 10^{-11} for 22 to 30 volts dc from a 26 volts reference and for 115/230 volts $\pm$ 10 percent.

d. DISTORTION.

The harmonic distortion shall be greater than 40 dB below rated output and the non-harmonic distortion shall be greater than 80 dB below rated output.

e. FREQUENCY ADJUSTMENTS.

(1) Fine adjustment. The fine frequency adjustment shall have a range of 5 x 10^{-8} minimum with a digital dial reading in parts in 10^{10}.

(2) Course adjustment. The course frequency adjustment shall have a range of 3 x 10^{-7} minimum.

(3) External adjustment. Application of a voltage which varies from +5 volts dc to -5 volts dc shall adjust the frequency by at least 2 x 10^{-8}.

CERTIFICATION.

The contractor shall certify that the product offered conforms to the requirements specified herein and to the producer's own drawings, specifications standards and quality assurance practices and is the same product that is offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract

PACKAGING AND MARKING FOR SHIPMENT.

The equipment shall be packaged and marked in accordance with standard commercial practice in order to preclude damage in shipment, unless otherwise specified by the procuring activity.

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KNOWN ACCEPTABLE PRODUCTS.

The following products are known to meet the requirements of this description.

Hewlett-Packard Model 105B, NSN 6625-00-405-0975.

Vectron Model FS 321A-1, NSN 6625-00-228-3182.

Austron Model 1250A, NSN 6625-01-035-6337.

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